

AI FDE End-to-End Production Delivery Spine

Global Workshop Execution Contract

Apply this to every stage:

"Analyse / Execute → Generate Required Artifacts → Save to Defined Repository Path → Cite Evidence → Record Assumptions & Unknowns → Validate Completion Gate → Proceed

Every generated artifact must:

- state stage, date/version, author/agent, status and evidence source;
- distinguish Verified Fact / Inference / Assumption / Unknown where relevant;
- reference actual repository paths, tests, schemas, logs, configurations or approved stakeholder evidence where applicable;
- include unresolved issues and residual risks;
- never claim completion when required evidence is missing;
- be version-controlled;
- not overwrite prior evidence silently;
- preserve earlier baselines so before/after comparison remains possible.

Discovery stages are read-only unless the prompt explicitly authorizes modification "

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AI FDE Stage	Final Workshop Prompt — Mandatory Artifact Generation Included	Primary Evidence / Artifacts
0A. Pre-Flight Repository & System Orientation	Establish a high-level factual orientation of the existing brownfield environment before changing anything. This stage is read-only. Do not modify application code, tests, dependencies, configuration, infrastructure, data or workflows. Inspect repository structure, technologies, applications, services, APIs, databases, integrations, ETL/data pipelines, tests, security mechanisms, observability, runtime/deployment clues and major business workflows. Identify what is	Repository/system orientation; technology inventory; high-level architecture; initial workflows/data/integration view; unknowns; risk register

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	clearly present, what appears operational, what is unknown and where deeper investigation is required. Do not perform deep refactoring analysis yet; that belongs to Stage 7. Distinguish Verified Fact, Inference, Assumption and Unknown and cite repository evidence. Create <code>docs/00-preflight/discovery/</code> and save <code>repository-overview.md</code> , <code>system-landscape.md</code> , <code>technology-inventory.md</code> , <code>high-level-architecture.md</code> , <code>integration-overview.md</code> , <code>data-flow-overview.md</code> , <code>workflow-overview.md</code> , <code>test-overview.md</code> , <code>security-observability-overview.md</code> , <code>assumptions-unknowns.md</code> , <code>initial-risk-register.md</code> and <code>discovery-summary.md</code> . Stage completion requires sufficient orientation to conduct engagement qualification without making transformation recommendations.	
0B. Provisional Operating Contract & Engineering Boundaries	Create a provisional engagement operating contract using currently available evidence. Because stakeholder discovery has not yet been completed, mark unresolved ownership or decision rights as PROVISIONAL rather than inventing them. Define current scope, exclusions, repositories, environments, data boundaries, permitted/prohibited actions, repository write restrictions,	Provisional operating contract; boundaries; provisional permissions; stop conditions; open governance decisions

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	production-access restrictions, security/privacy constraints, provisional tool and agent permissions, evidence requirements, change-control expectations, human-approval triggers, stop conditions and escalation assumptions. Create <pre>docs/00-preflight/operating-contract/</pre> and save <pre>provisional-operating-contract.md , scope-boundaries.md , repository-write-boundaries.md , environment-access-boundaries.md , data-use-constraints.md , provisional-tool-agent-permissions.md , provisional-human-approval-rules.md , evidence-contract.md , change-control-rules.md , stop-conditions.md , open-governance-decisions.md operating-contract-readiness.md</pre> and Explicitly flag items requiring confirmation during Stages 2, 23, 25 and 37. Do not treat provisional decisions as final authority.	
0C. Provisional Token Efficiency & AI Economics Envelope	Establish a provisional economic and resource envelope without assuming that AI has already been justified. Create scenarios for deterministic software, conventional automation, GenAI and agentic AI where relevant. Estimate likely workload volume, context size, model calls, input/output/retrieval/tool tokens, agent-loop overhead, retries,	Provisional token/context/cost envelope; solution scenarios; workload assumptions; preliminary TCO

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	latency and infrastructure consumption only where enough evidence exists. Clearly separate measured data from assumptions. Identify maximum acceptable cost/request, cost/case, latency, context size and agent-loop limits as provisional design constraints. Create <code>docs/00-preflight/ai-economics/</code> and save <code>workload-assumptions.md</code> , <code>solution-scenarios.md</code> , <code>provisional-model-call-inventory.md</code> , <code>token-flow-scenarios.md</code> , <code>provisional-token-budget.md</code> , <code>provisional-context-budget.md</code> , <code>provisional-loop-budget.md</code> , <code>volume-assumptions.md</code> , <code>cost-scenarios.md</code> , <code>quality-latency-cost-envelope.md</code> , <code>preliminary-finops-baseline.md</code> , <code>preliminary-tco.md</code> and <code>economics-readiness.md</code> . Stage 8 determines whether the AI-specific portions remain applicable.	
1. Engage & Qualify	Qualify the engagement using Stage 0 evidence. Identify client context, business problem, sponsor, business owner, technical owner, urgency, expected outcomes, dependencies, delivery constraints and reasons the initiative could be premature or non-viable. Separate stakeholder claims from validated evidence. Create <code>docs/01-engagement/</code> and save	Engagement Canvas; Team Charter; qualification evidence; Go/No-Go

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	engagement-canvas.md , team-charter.md , qualification-checklist.md , use-case-hypothesis.md , engagement-risks.md , open-qualification-questions.md and engagement-go-no-go.md . The final artifact must classify the engagement as Go, Conditional Go or No-Go with evidence.	
2. Stakeholder Discovery & Authority Confirmation	Identify stakeholders across business, users, product, architecture, engineering, data, security, privacy, compliance, risk, operations and leadership. Capture needs, incentives, concerns, responsibilities, conflicts, approval authority and decision rights. Reconcile these findings with Stage 0B and update provisional governance assumptions where evidence permits. Create docs/02-stakeholders/ and save stakeholder-map.md , stakeholder-needs-matrix.md , decision-rights-map.md , approval-authority-map.md , stakeholder-conflicts.md , escalation-map.md and operating-contract-confirmations.md . Do not invent owners where none have been assigned; record them as unresolved governance gaps.	Stakeholder map; needs; decision rights; approval authority; operating-contract confirmations
3. Frame Problem & Value	Translate stakeholder pain into a precise business problem independent of any preferred technology. Separate symptom, root problem, proposed solution and assumed AI need. Define personas, jobs-to-be-	Problem framing; personas/ JTBD; requirements; NFRs; success criteria; value hypothesis

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	done, journeys, scope, exclusions, requirements, NFRs and measurable success/failure criteria. Create <code>docs/03-problem-value/</code> and save <code>problem-framing-canvas.md</code> , <code>problem-statement.md</code> , <code>personas.md</code> , <code>jobs-to-be-done.md</code> , <code>user-journeys.md</code> , <code>scope-exclusions.md</code> , <code>business-requirements.md</code> , <code>nfrs.md</code> , <code>success-criteria.md</code> , <code>value-hypothesis.md</code> and <code>assumptions-register.md</code> . Stage completion requires a technology-neutral problem definition.	
4. Capture Before-AI / Before-Intervention KPIs	Establish the current-state measurement baseline before any intervention. Define KPI name, meaning, formula, unit, owner, source, measurement period, segmentation and current value. Include cycle time, effort, error, rework, quality, throughput, leakage, compliance, cost, adoption and relevant domain metrics. Validate measurement quality and define proxies where evidence is weak. Freeze metric definitions for later comparison. Create <code>docs/04-baseline-kpis/</code> and save <code>kpi-dictionary.md</code> , <code>baseline-kpi-sheet.md</code> , <code>measurement-methodology.md</code> , <code>data-source-map.md</code> , <code>baseline-data-quality.md</code> , <code>proxy-metrics.md</code> ,	Frozen KPI baseline; KPI dictionary; measurement methodology; evidence

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	<code>baseline-evidence.md</code> and <code>kpi-baseline-signoff.md</code> .	
5. Discover Current Process & Brownfield Landscape	Map the current end-to-end business process and system landscape. Identify actors, steps, decisions, queues, handoffs, systems, APIs, events, databases, ETL, batch jobs, manual workarounds, integrations, controls and operational dependencies. Highlight bottlenecks, rework, duplication, delays, control gaps and fragile dependencies. Create <code>docs/05-current-state/</code> and save <code>current-process-map.md</code> , <code>business-workflow-map.md</code> , <code>system-landscape.md</code> , <code>integration-landscape.md</code> , <code>brownfield-inventory.md</code> , <code>manual-handoffs.md</code> , <code>process-bottlenecks.md</code> , <code>control-points.md</code> , <code>dependency-hotspots.md</code> and <code>current-state-summary.md</code> .	Current process/system maps; brownfield inventory; bottleneck and dependency analysis
6. Root-Cause Analysis	Perform evidence-based causal analysis across process, people, policy, data, technology, organization and architecture. Use 5-Whys, Fishbone, causal trees or equivalent techniques. For every proposed cause record supporting evidence, contradictory evidence, confidence and validation method. Never label correlation as causation without evidence. Create <code>docs/06-root-cause/</code> and save <code>root-cause-tree.md</code> , <code>five-whys.md</code> , <code>fishbone-analysis.md</code> , <code>causal-hypotheses.md</code> , <code>evidence-confidence-matrix.md</code> ,	Root-cause analysis; evidence-confidence matrix; validated causal hypotheses

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	<code>validation-plan.md</code> , <code>confirmed-root-causes.md</code> and <code>root-cause-readiness.md</code> .	
7. Deep Brownfield Repository Assessment & Behaviour Baseline	Perform the detailed engineering assessment that was intentionally deferred from Stage 0A. Do not change intended behaviour. Analyse architecture drift, module boundaries, dependencies, duplication, complexity, obsolete libraries, legacy patterns, partial migrations, configuration, data handling, security, testability, observability and maintainability. Run existing tests where safe. Identify critical workflows whose behaviour must be preserved. Where test coverage is insufficient, define characterization tests before refactoring. Create <code>docs/07-repo-assessment/</code> and save <code>repo-assessment.md</code> , <code>architecture-drift.md</code> , <code>dependency-map.md</code> , <code>technical-debt-register.md</code> , <code>legacy-pattern-register.md</code> , <code>partial-migration-register.md</code> , <code>critical-workflow-inventory.md</code> , <code>baseline-test-results.md</code> , <code>baseline-behaviour.md</code> , <code>characterization-test-plan.md</code> , <code>security-code-quality-findings.md</code> , <code>maintainability-assessment.md</code> and <code>repo-transformation-readiness.md</code> . Do not refactor during this stage.	Deep repo assessment; behaviour baseline; technical debt; characterization plan
8. AI-vs-No-AI Qualification	Evaluate each intervention against process redesign, policy change, deterministic	Intervention decision; AI justification; minimum

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	software, rules engines, workflow automation, analytics, classical ML, GenAI and agentic AI. Apply minimum necessary intelligence and minimum necessary agency. Explicitly explain what uncertainty, language, reasoning, generation or adaptive decision requirement justifies AI. Reject AI where conventional approaches are safer, cheaper or more reliable. Reconcile the result with Stage 0C and retire irrelevant AI cost assumptions. Create <code>docs/08-ai-qualification/</code> and save <code>intervention-options.md</code> , <code>ai-vs-no-ai-matrix.md</code> , <code>deterministic-vs-ai-boundaries.md</code> , <code>minimum-intelligence-assessment.md</code> , <code>minimum-agency-assessment.md</code> , <code>intervention-hypothesis.md</code> , <code>economics-baseline-reconciliation.md</code> and <code>qualification-decision.md</code> .	intelligence/agency; economics reconciliation
9. Initial PRD / Product Intent	Create the Initial PRD from validated discovery and qualification. Define problem, personas, journeys, capabilities, functional requirements, NFRs, exclusions, constraints, dependencies, success metrics, risks, MVP scope, future scope and unresolved questions. Trace requirements to business problems and evidence. Create <code>docs/09-initial-prd/</code> and save <code>initial-prd.md</code> , <code>capability-map.md</code> , <code>functional-requirements.md</code> ,	Initial PRD; capability map; requirements; MVP/non-goals; traceability

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	nfrs.md , mvp-scope.md , non-goals.md , release-hypothesis.md , product-risks.md , open-product-decisions.md and initial-prd-traceability.md .	
10. Target Architecture + ADRs	Develop and compare architecture options across application, frontend/backend, APIs, AI/model, data, knowledge, integration, security, identity, observability, human control and deployment. Evaluate them against functional requirements, NFRs, security, portability, cost, latency, scalability, resilience and maintainability. Create docs/10-architecture/ and docs/10-architecture/adrs/ . Save architecture-options.md , target-architecture.md , component-model.md , integration-architecture.md , deployment-architecture.md , security-architecture.md , model-service-scorecard.md , architecture-risk-register.md , architecture-decision-summary.md and one ADR per material decision.	Target architecture; alternatives; ADRs; model/service scorecard
11. Data, Knowledge & Context Strategy	Define data, knowledge and runtime-context architecture. Cover sources, ownership, schemas, contracts, quality, lineage, provenance, freshness, retention, metadata, PII and access controls. Define ingestion, chunking, indexing, retrieval, reranking, grounding, context assembly,	Data contracts; knowledge model; lineage; retrieval/context strategy; synthetic-data plan

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	compaction and provenance/citation requirements. Define synthetic-data requirements where production data cannot be used. Create <code>docs/11-data-context/</code> and save <code>data-source-inventory.md</code> , <code>data-contracts.md</code> , <code>data-quality-rules.md</code> , <code>lineage-map.md</code> , <code>knowledge-model.md</code> , <code>metadata-model.md</code> , <code>retrieval-strategy.md</code> , <code>context-engineering-strategy.md</code> , <code>context-compaction-policy.md</code> , <code>provenance-policy.md</code> , <code>synthetic-data-strategy.md</code> and <code>data-context-risks.md</code> .	
12. Spec-Driven Design	Convert approved product intent and architecture into implementation-ready specifications before coding. Define feature/system behaviour, APIs, events, schemas, state transitions, business rules, errors, constraints, security, observability, NFRs and acceptance criteria. Create <code>docs/12-specs/</code> and <code>docs/12-specs/features/</code> . Save <code>system-spec.md</code> , feature-specific specifications, <code>api-contracts.md</code> , <code>event-contracts.md</code> , <code>schema-specifications.md</code> , <code>interface-contracts.md</code> , <code>business-rules.md</code> , <code>error-handling-spec.md</code> , <code>security-spec.md</code> , <code>observability-spec.md</code> , <code>nfr-spec.md</code> , <code>acceptance-criteria.md</code> and	System/feature specs; API/ event/schema contracts; NFR/security/observability specs

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	<code>spec-readiness.md</code> . Do not implement material ambiguity.	
13. Requirements → Specs → Tests → Evidence Traceability	Build bidirectional traceability for functional, data, AI-quality, security, privacy, resilience, performance, observability and cost requirements. Map requirement → spec → planned implementation → test/evaluation → evidence. Detect orphan requirements, untested specs, unjustified implementation and unverifiable criteria. Create <code>docs/13-traceability/</code> and save <code>requirements-traceability-matrix.md</code> , <code>spec-to-test-map.md</code> , <code>test-to-evidence-map.md</code> , <code>security-traceability.md</code> , <code>data-traceability.md</code> , <code>ai-quality-traceability.md</code> , <code>resilience-traceability.md</code> , <code>cost-traceability.md</code> , <code>coverage-gap-register.md</code> and <code>traceability-summary.md</code> . Critical unexplained gaps block progression.	End-to-end traceability; evidence mapping; coverage gaps
14. Transformation / Migration Plan	Design the transition from brownfield state to target state in safe increments. Define dependencies, sequencing, coexistence, adapters, backward compatibility, feature flags, schema/data migration, cutover, rollback and verification. Explicitly identify components that should remain unchanged. Create <code>docs/14-transformation/</code> and save <code>transformation-backlog.md</code> , <code>migration-roadmap.md</code> ,	Transformation backlog; migration/cutover/rollback strategy

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	<code>dependency-sequence.md</code> , <code>coexistence-strategy.md</code> , <code>adapter-strategy.md</code> , <code>schema-migration-plan.md</code> , <code>feature-flag-plan.md</code> , <code>cutover-plan.md</code> , <code>rollback-strategy.md</code> , <code>transformation-risks.md</code> and <code>transformation-gates.md</code> .	
15. Transform Bad Repo → Good Repo	Execute only the approved transformation. Preserve validated behaviour except where approved specifications intentionally change it. Improve repository structure, boundaries, dependencies, configuration, tests, maintainability, documentation, security and engineering automation in small reviewable increments. Continuously run relevant tests. Create <code>docs/15-modernization/</code> and save <code>transformation-log.md</code> , <code>changed-components.md</code> , <code>behavior-preservation-evidence.md</code> , <code>approved-behavior-changes.md</code> , <code>refactoring-decisions.md</code> , <code>dependency-changes.md</code> , <code>remaining-technical-debt.md</code> , <code>test-execution-log.md</code> and <code>modernization-summary.md</code> . Modify implementation only within approved Stage 0B boundaries.	Modernized repo; transformation log; behavior-preservation evidence
16. Validate the Good Repo	Validate the transformed repository against target architecture, specifications and engineering standards. Run unit, integration, contract, characterization, security and data-contract tests plus relevant static/ dependency/architecture	Quality gate; regression/ conformance/security evidence; validation sign-off

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	checks. Compare Stage 7 baseline behaviour with transformed behaviour and explain every difference. Create <code>docs/16-repo-validation/</code> and save <code>repo-quality-gate.md</code> , <code>regression-results.md</code> , <code>architecture-conformance-report.md</code> , <code>test-results.md</code> , <code>security-scan-summary.md</code> , <code>dependency-scan-summary.md</code> , <code>data-contract-results.md</code> , <code>behavior-difference-report.md</code> , <code>remaining-debt-register.md</code> and <code>validation-signoff.md</code> .	
17. Finalise Implementation PRD	Reconcile the Initial PRD with repository findings, architecture decisions and transformation evidence. Update capabilities, priorities, scope, dependencies, NFRs, acceptance criteria and release boundaries. Record every material change and invalidated assumption. Create <code>docs/17-implementation-prd/</code> and save <code>implementation-prd.md</code> , <code>prd-change-log.md</code> , <code>prioritized-capabilities.md</code> , <code>finalized-nfrs.md</code> , <code>release-boundaries.md</code> , <code>finalized-acceptance-criteria.md</code> , <code>invalidated-assumptions.md</code> and <code>implementation-baseline-signoff.md</code> . This becomes the implementation baseline, not the final as-built PRD.	Implementation PRD; change log; release/NFR/ acceptance baseline

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18. Evidence-Driven Delivery Planning	Translate the Implementation PRD and specifications into executable increments. Define stories/tasks, dependencies, owners, tests, evaluation datasets, acceptance conditions, Definition of Done and evidence requirements. Create <code>docs/18-delivery/</code> and save <code>delivery-backlog.md</code> , <code>increment-plan.md</code> , <code>ownership-map.md</code> , <code>definition-of-done.md</code> , <code>acceptance-test-plan.md</code> , <code>evaluation-rubric.md</code> , <code>evidence-checklist.md</code> , <code>delivery-dependencies.md</code> and <code>delivery-risk-register.md</code> . No work item is complete without its required evidence.	Delivery backlog; DoD; evaluation rubric; evidence checklist
19. Intelligence Core — Prompt, Retrieval & Model Engineering	Implement the approved intelligence layer: version-controlled prompts, model configuration, structured outputs, extraction, embeddings, retrieval, reranking, grounding and inference. Build evaluation datasets before tuning. Benchmark models where justified. Measure retrieval relevance, groundedness, answer quality, hallucination/unsupported claims, abstention, latency and cost. Create <code>docs/19-intelligence/</code> and save <code>prompt-registry.md</code> , <code>model-configuration.md</code> , <code>model-benchmark.md</code> , <code>extraction-pipeline.md</code> , <code>retrieval-pipeline.md</code> ,	Prompt/model/RAG assets; evaluation datasets/results; intelligence gate

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	<code>rag-design.md</code> , <code>evaluation-dataset-spec.md</code> , <code>rag-evaluation-results.md</code> , <code>prompt-evaluation-results.md</code> , <code>grounding-results.md</code> , <code>quality-latency-cost-results.md</code> and <code>intelligence-release-gate.md</code> . Store executable prompts/ configuration in appropriate version-controlled implementation directories as well.	
20. Application & Services Engineering	Implement UI, APIs, business logic, persistence and application services according to specifications. Keep deterministic business logic separate from probabilistic AI behaviour. Implement validation, errors, authentication, authorization, idempotency, accessibility, auditability and observability hooks. Create <code>docs/20-application/</code> and save <code>application-component-map.md</code> , <code>api-implementation-map.md</code> , <code>state-model.md</code> , <code>business-logic-map.md</code> , <code>deterministic-ai-boundary.md</code> , <code>error-handling-implementation.md</code> , <code>access-control-implementation.md</code> , <code>application-test-summary.md</code> and <code>application-readiness.md</code> .	Application/services; deterministic-AI boundary; tests/readiness
21. Enterprise Integration	Integrate systems of record, legacy applications, APIs, databases, events and enterprise workflows. Validate authentication, authorization, schemas, contracts, retries, idempotency, reconciliation, timeout behaviour, stale	Integration inventory/ contracts; adapters; test/ reconciliation evidence

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	data, duplicates and dependency failures. Create <code>docs/21-integration/</code> and save <code>integration-inventory.md</code> , <code>api-contracts.md</code> , <code>data-contracts.md</code> , <code>event-contracts.md</code> , <code>adapter-map.md</code> , <code>authentication-map.md</code> , <code>failure-semantics.md</code> , <code>reconciliation-rules.md</code> , <code>integration-test-results.md</code> and <code>integration-readiness.md</code> .	
22. Agent, Tool, Harness, Loop & Graph Engineering	Only where Stage 8 justified agentic behaviour, engineer the full agent runtime. Define planner/reasoner roles, strict tool schemas, tool gateways, state, memory, context management, workflow graph, loop controls, termination criteria, token/time/tool budgets, retries, evaluator/critic logic, sandboxing, permissions and recovery. Prevent uncontrolled recursion, privilege expansion, tool misuse and context accumulation. Create <code>docs/22-agentic-engineering/</code> and save <code>agent-architecture.md</code> , <code>harness-specification.md</code> , <code>tool-contracts.md</code> , <code>tool-permission-map.md</code> , <code>state-model.md</code> , <code>memory-policy.md</code> , <code>context-policy.md</code> , <code>graph-definition.md</code> , <code>loop-control-policy.md</code> , <code>termination-rules.md</code> , <code>agent-budget-policy.md</code> ,	Agent/harness/tool/graph/loop assets; test evidence or N/A justification

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	<code>sandbox-policy.md</code> , <code>agent-test-plan.md</code> , <code>agent-test-results.md</code> and <code>agent-readiness-gate.md</code> . If agentic AI was rejected at Stage 8, create <code>not-applicable.md</code> with justification rather than inventing agent artifacts.	
23. HITL/HOTL & Deterministic Control Boundaries	Classify material decisions/ actions by autonomy level. Define what remains deterministic, what AI may recommend, what AI may execute after validation, what requires explicit human approval and what must never be delegated. Reconcile these decisions with Stage 0B provisional controls. Create <code>docs/23-human-control/</code> and save <code>autonomy-matrix.md</code> , <code>deterministic-control-rules.md</code> , <code>hitl-workflow.md</code> , <code>hotl-workflow.md</code> , <code>approval-gates.md</code> , <code>human-override-policy.md</code> , <code>confidence-handling.md</code> , <code>escalation-matrix.md</code> , <code>operating-contract-updates.md</code> and <code>human-control-test-results.md</code> .	Autonomy matrix; human-control workflows; approval/override rules
24. Security + Privacy + Responsible AI by Design	Threat-model application, APIs, identity, data, prompts, RAG, models, agents, tools, context, supply chain and integrations. Address prompt/indirect injection, data exfiltration, insecure output handling, excessive agency, insecure tool use, poisoning, secrets exposure, authorization bypass, privacy/PII and relevant responsible-AI risks. Define	Threat model; security/privacy/RAI controls; initial control tests

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	and implement controls within approved scope. Create <code>docs/24-security-privacy/</code> and save <code>threat-model.md</code> , <code>attack-surface-map.md</code> , <code>security-control-matrix.md</code> , <code>privacy-impact-assessment.md</code> , <code>prompt-injection-controls.md</code> , <code>agent-security-controls.md</code> , <code>data-protection-controls.md</code> , <code>responsible-ai-controls.md</code> , <code>security-test-plan.md</code> , <code>security-test-results.md</code> , <code>residual-security-risks.md</code> and <code>security-readiness.md</code> .	
25. Governance + Compliance + Assurance	Map applicable regulatory, contractual, organizational and AI-governance obligations to controls, owners, evidence and approval points. Confirm or update Stage 0B governance assumptions. Define accountability, registration, change control, risk acceptance, auditability and evidence retention. Create <code>docs/25-governance/</code> and save <code>compliance-obligations.md</code> , <code>compliance-control-matrix.md</code> , <code>governance-model.md</code> , <code>accountability-map.md</code> , <code>ai-system-register.md</code> , <code>model-card.md</code> , <code>system-card.md</code> , <code>risk-acceptance-register.md</code> , <code>evidence-retention-policy.md</code> , <code>audit-evidence-index.md</code> , <code>operating-contract-governance-updates.md</code> and <code>governance-readiness.md</code> .	Compliance/governance package; AI registry; model/system cards; assurance evidence
26. TEVV + AI Red Teaming	Execute formal Test, Evaluation, Verification and Validation across deterministic and	TEVV datasets/results; red-team findings; release gate

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	probabilistic behaviour. Test representative, golden, edge, adversarial, safety and failure datasets. Evaluate functionality, model/RAG quality, groundedness, hallucination, abstention, agents/tools where applicable, security, privacy, resilience and business outcomes. Red-team prompt injection, data leakage, control bypass, excessive agency and tool abuse. Create <code>docs/26-tevv/</code> and save <code>tevv-plan.md</code>, <code>golden-dataset-spec.md</code>, <code>edge-dataset-spec.md</code>, <code>adversarial-dataset-spec.md</code>, <code>failure-dataset-spec.md</code>, <code>evaluation-metrics.md</code>, <code>release-thresholds.md</code>, <code>tevv-results.md</code>, <code>red-team-plan.md</code>, <code>red-team-findings.md</code>, <code>remediation-evidence.md</code>, <code>residual-tevv-risks.md</code> and <code>tevv-release-gate.md</code>. Release thresholds must be objective and predeclared where feasible.	
27. Production Hardening	Harden identity, RBAC, secrets, network/runtime, infrastructure, dependencies, APIs, CI/CD, build artifacts and software supply chain. Generate/review SBOMs and scan source, dependencies, infrastructure and artifacts. Remove debug behavior, test secrets, unnecessary ports, excessive privileges and insecure defaults. Create <code>docs/27-hardening/</code> and save <code>hardening-plan.md</code>, <code>identity-rbac-hardening.md</code>,	Hardened production baseline; SBOM/scans; vulnerability disposition

AI FDE Stage	Final Workshop Prompt — Mandatory Artifact Generation Included	Primary Evidence / Artifacts
	<code>secrets-hardening.md</code> , <code>runtime-hardening.md</code> , <code>dependency-hardening.md</code> , <code>supply-chain-controls.md</code> , <code>sbom-summary.md</code> , <code>vulnerability-register.md</code> , <code>scan-results.md</code> , <code>vulnerability-disposition.md</code> <code>production-readiness-checklist.md</code> .	
28. Reliability & Resilience Engineering	Design and test realistic failures. Define timeouts, retries, backoff, circuit breakers, bulkheads, fallback models/providers, caching, capacity limits, graceful degradation and AI-disabled operation. Prevent retry storms and cascading agent/tool failure. Create <code>docs/28-resilience/</code> and save <code>resilience-architecture.md</code> , <code>failure-mode-analysis.md</code> , <code>timeout-retry-policy.md</code> , <code>circuit-breaker-policy.md</code> , <code>fallback-matrix.md</code> , <code>capacity-plan.md</code> , <code>graceful-degradation-design.md</code> , <code>ai-disabled-mode.md</code> , <code>failure-injection-plan.md</code> , <code>failure-injection-results.md</code> and <code>resilience-readiness.md</code> .	Resilience architecture; capacity/failure evidence; degraded modes
29. AI Incident Response + BC/DR	Define incident and continuity procedures for prompt injection, unsafe actions, compromised tools, data leakage, bad retrieval, model/provider failure, cost runaway, corrupted configuration, infrastructure outages and disaster scenarios. Define detection, severity, containment, evidence preservation, communications, recovery,	Incident playbook; BC/DR; recovery/tabletop evidence

AI FDE Stage	Final Workshop Prompt — Mandatory Artifact Generation Included	Primary Evidence / Artifacts
	RTO/RPO and failover. Conduct a tabletop or simulation. Create <code>docs/29-incident-bcdr/</code> and save <code>ai-incident-playbook.md</code> , <code>incident-severity-matrix.md</code> , <code>containment-procedures.md</code> , <code>forensics-evidence-policy.md</code> , <code>communications-plan.md</code> , <code>bc-plan.md</code> , <code>dr-plan.md</code> , <code>rto-rpo.md</code> , <code>failover-procedure.md</code> , <code>recovery-procedure.md</code> , <code>tabletop-scenario.md</code> , <code>tabletop-results.md</code> and <code>incident-readiness.md</code> .	
30. Release / Cutover / Rollback	Prepare a controlled production release. Define environment promotion, approvals, deployment method, feature flags, canary/phased release where appropriate, migration, validation, monitoring, go/no-go criteria and rollback triggers. Verify backup and rollback before cutover. Create <code>docs/30-release/</code> and save <code>release-plan.md</code> , <code>environment-promotion-plan.md</code> , <code>deployment-strategy.md</code> , <code>feature-flag-plan.md</code> , <code>cutover-checklist.md</code> , <code>go-no-go-criteria.md</code> , <code>rollback-plan.md</code> , <code>backup-validation.md</code> , <code>release-approvals.md</code> , <code>deployment-evidence.md</code> and <code>release-outcome.md</code> .	Release/cutover/rollback evidence
31. Observability + SLO/SLA Engineering	Instrument application, infrastructure, data, RAG, models and agents where applicable. Capture privacy-safe logs, metrics and traces for requests, retrieval,	Logs/metrics/traces; AI/agent telemetry; SLO/SLA; runbook

AI FDE Stage	Final Workshop Prompt — Mandatory Artifact Generation Included	Primary Evidence / Artifacts
	prompt/config versions, model latency, tokens, tool calls, agent trajectories, errors and business outcomes. Define SLOs, SLAs, alerts and error budgets. Create <code>docs/31-observability/</code> and save <code>observability-architecture.md</code> , <code>logging-spec.md</code> , <code>metrics-spec.md</code> , <code>tracing-spec.md</code> , <code>ai-telemetry-spec.md</code> , <code>agent-telemetry-spec.md</code> , <code>dashboard-catalog.md</code> , <code>alert-catalog.md</code> , <code>slo-sla-definitions.md</code> , <code>error-budget-policy.md</code> , <code>operational-runbook.md</code> and <code>observability-validation.md</code> .	
32. Token Efficiency + AI FinOps + TCO	Measure actual production economics against the applicable Stage 0C scenario. Analyse input/output/context/retrieval/tool/agent-loop tokens, model calls, cache hits, retries, latency, throughput, infrastructure cost and human-oversight cost. Calculate unit economics per request, case, workflow, user and business outcome where meaningful. Identify token leakage and value leakage. Optimise without violating quality thresholds. Create <code>docs/32-finops/</code> and save <code>production-token-dashboard-spec.md</code> , <code>baseline-vs-actual-token-analysis.md</code> , <code>model-usage-analysis.md</code> , <code>cache-effectiveness.md</code> , <code>cost-per-request.md</code> , <code>cost-per-case.md</code> , <code>cost-per-workflow.md</code> ,	Actual-vs-baseline economics; FinOps/TCO; optimization backlog

AI FDE Stage	Final Workshop Prompt — Mandatory Artifact Generation Included	Primary Evidence / Artifacts
	<code>cost-per-outcome.md</code> , <code>token-leakage-analysis.md</code> , <code>optimization-backlog.md</code> , <code>finops-model.md</code> , <code>tco-model.md</code> and <code>cost-guardrails.md</code> . If Stage 8 rejected AI, adapt this stage to actual platform/ infrastructure economics instead of inventing token metrics.	
33. Vendor / Third-Party / Concentration Risk	Assess critical model, platform, infrastructure, data and service providers. Evaluate availability, contractual terms, security/ privacy, data usage, regulatory constraints, price risk, portability, concentration, lock-in, substitution and exit feasibility. Create <code>docs/33-vendor-risk/</code> and save <code>vendor-inventory.md</code> , <code>vendor-scorecards.md</code> , <code>third-party-risk-assessment.md</code> , <code>data-processing-dependencies.md</code> , <code>concentration-risk-analysis.md</code> , <code>lock-in-analysis.md</code> , <code>portability-assessment.md</code> , <code>substitution-strategy.md</code> , <code>exit-plan.md</code> and <code>vendor-risk-acceptance.md</code> .	Vendor/third-party assessment; concentration and exit strategy
34. Measure After-Intervention KPIs	Measure production performance using the metric definitions frozen in Stage 4. Include successes, failures, exceptions, overrides and adoption. Confirm measurement-window and population comparability before making claims. Create <code>docs/34-after-kpis/</code> and save	After-intervention KPI evidence; adoption/ exception metrics; comparability check

AI FDE Stage	Final Workshop Prompt — Mandatory Artifact Generation Included	Primary Evidence / Artifacts
	after-intervention-kpi-sheet.md , production-performance-report.md , adoption-metrics.md , exception-metrics.md , human-override-metrics.md , measurement-data-quality.md and kpi-comparability-check.md . Do not change KPI definitions to improve results.	
35. Before-vs-After Variance + Value Leakage	Compare Stage 4 and Stage 34 using frozen definitions and comparable populations. Quantify improvement, deterioration and deviation from expected benefit. Identify leakage from low adoption, overrides, errors, rework, latency, control friction, AI/ infrastructure cost or workflow displacement. Avoid attributing all change to the intervention where confounders exist. Create docs/35-value-leakage/ and save before-after-analysis.md , kpi-variance-table.md , benefit-variance.md , value-leakage-analysis.md , confounder-analysis.md , causal-findings.md , adoption-leakage.md , cost-leakage.md and corrective-action-backlog.md .	KPI variance; value leakage; confounder analysis; corrective actions
36. ROI / NPV / Benefits Realisation	Translate only verified improvements into financial value. Include adoption, productivity, quality, risk, revenue/cost effects, model/ token/infrastructure costs and human oversight. Avoid double-counting benefits	Verified ROI/NPV/payback; sensitivity analysis; benefits sign-off

AI FDE Stage	Final Workshop Prompt — Mandatory Artifact Generation Included	Primary Evidence / Artifacts
	already captured in other metrics. Test optimistic, expected and downside scenarios. Create <code>docs/36-benefits/</code> and save <code>benefit-assumptions.md</code> , <code>verified-benefits.md</code> , <code>cost-model.md</code> , <code>roi-model.md</code> , <code>npv-model.md</code> , <code>payback-analysis.md</code> , <code>scenario-analysis.md</code> , <code>sensitivity-analysis.md</code> , <code>double-counting-check.md</code> , <code>benefits-realisation-dashboard-spec.md</code> and <code>benefits-signoff.md</code> .	
37. Target Operating Model + RACI + Ownership	Convert the temporary Stage 0B engagement contract into steady-state production ownership. Assign responsibility for application, data, models, prompts, retrieval, agents/tools where applicable, security, incidents, observability, costs, vendors, governance and business outcomes. Define support tiers, change approval, service ownership, escalation and decision rights. Create <code>docs/37-operating-model/</code> and save <code>target-operating-model.md</code> , <code>raci.md</code> , <code>service-ownership-map.md</code> , <code>model-ownership.md</code> , <code>data-ownership.md</code> , <code>agent-tool-ownership.md</code> , <code>security-ownership.md</code> , <code>finops-ownership.md</code> , <code>support-model.md</code> , <code>change-governance.md</code> , <code>decision-rights.md</code> , <code>escalation-model.md</code> and <code>operating-contract-closure.md</code> .	Production operating model; RACI; ownership; support/change governance

AI FDE Stage	Final Workshop Prompt — Mandatory Artifact Generation Included	Primary Evidence / Artifacts
38. Handover + Knowledge Transfer	Transfer architecture, code, configuration, prompts, model settings, data/ knowledge pipelines, agent workflows where applicable, operational procedures, security, governance and runbooks. Use walkthroughs plus practical operator exercises. Confirm the receiving team can deploy, observe, diagnose, rollback and recover the solution independently. Create <code>docs/38-handover/</code> and save <code>handover-index.md</code> , <code>knowledge-transfer-plan.md</code> , <code>architecture-handover.md</code> , <code>engineering-handover.md</code> , <code>ai-rag-agent-handover.md</code> , <code>security-handover.md</code> , <code>operations-handover.md</code> , <code>operator-exercises.md</code> , <code>operator-exercise-results.md</code> , <code>open-handover-items.md</code> and <code>handover-acceptance-signoff.md</code> .	Handover pack; operator exercises; acceptance sign-off
39. Drift + Continuous Improvement	Define continuous evaluation across data, knowledge, retrieval, prompts, models, agents, application quality, adoption, reliability, latency, tokens/ cost and changing requirements as applicable. Define thresholds that trigger investigation or controlled change. Do not permit untracked prompt/ model changes in production. Create <code>docs/39-continuous-improvement/</code> and save <code>drift-framework.md</code> , <code>data-drift-metrics.md</code> , <code>knowledge-drift-metrics.md</code> , <code>retrieval-drift-metrics.md</code> ,	Drift/evaluation framework; change triggers; improvement backlog

AI FDE Stage	Final Workshop Prompt — Mandatory Artifact Generation Included	Primary Evidence / Artifacts
	prompt-model-drift-metrics.md , agent-drift-metrics.md , evaluation-cadence.md , change-trigger-thresholds.md , continuous-evaluation-plan.md , controlled-change-process.md and improvement-backlog.md .	
40. Scale / 90-Day Transformation Roadmap	Identify only capabilities, components, prompts, controls, contracts, evaluation assets and architecture patterns that have demonstrated sufficient evidence for reuse. Assess adjacent use cases for value, readiness, commonality, dependencies and risk. Prevent copying first-use-case assumptions or debt. Create docs/40-scale/ and save reusable-components.md , pattern-catalog.md , accelerator-pack.md , adjacent-use-case-inventory.md , reuse-readiness-assessment.md , scale-risk-assessment.md , scale-out-backlog.md , 90-day-roadmap.md and scale-governance.md .	Validated reusable assets; scale readiness; 90-day roadmap
41. Retirement / Decommissioning	Define safe retirement criteria for models, prompts, indexes, APIs, agents/tools, applications, databases, infrastructure and vendors. Address retention, legal hold, archival, dependency removal, access/key revocation, migration, contracts and audit evidence. Confirm no live dependency remains before deletion. Create docs/41-retirement/ and save retirement-criteria.md ,	Retirement/decommission criteria; dependency clearance; revocation/archive evidence

AI FDE Stage	Final Workshop Prompt — Mandatory Artifact Generation Included	Primary Evidence / Artifacts
	<code>dependency-clearance.md</code> , <code>data-retention-plan.md</code> , <code>archive-plan.md</code> , <code>legal-hold-check.md</code> , <code>access-revocation-plan.md</code> , <code>secret-key-revocation.md</code> , <code>vendor-termination-plan.md</code> , <code>decommission-checklist.md</code> and <code>retirement-evidence.md</code> .	
42. Executive Defence & Final Value Story	Create the final evidence-backed executive narrative. Explain the original problem and baseline, validated root causes, intervention choice, why AI was or was not appropriate, architecture and ADRs, brownfield transformation, intelligence/agent engineering where applicable, control framework, TEVV evidence, production outcomes, economics, realised benefits, failures, trade-offs, remaining risks and next steps. Every material claim must map to lifecycle evidence. Create <code>docs/42-executive/</code> and save <code>executive-narrative.md</code> , <code>problem-to-value-story.md</code> , <code>architecture-defence.md</code> , <code>ai-decision-defence.md</code> , <code>risk-control-summary.md</code> , <code>tevv-summary.md</code> , <code>production-outcomes.md</code> , <code>economic-value-summary.md</code> , <code>lessons-learned.md</code> , <code>residual-risks.md</code> , <code>next-step-recommendations.md</code> , <code>demo-day-script.md</code> , <code>elevator-pitch.md</code> and <code>executive-evidence-index.md</code> .	Executive narrative; evidence index; architecture/value defence; lessons/next steps
FINAL. Authoritative Final As-Built PRD	Create the authoritative product baseline representing what was actually delivered, validated,	Final As-Built PRD; requirement disposition; as-built architecture; final

AI FDE Stage	Final Workshop Prompt — Mandatory Artifact Generation Included	Primary Evidence / Artifacts
	released and accepted. Reconcile Stage 9 Initial PRD, Stage 17 Implementation PRD, system/feature specs, ADRs, implementation, production configuration, TEVV evidence, data/ knowledge architecture, AI/ model/RAG/agent behaviour where applicable, integrations, deterministic/ HITL boundaries, security/ privacy controls, governance obligations, SLO/SLA commitments, operating model, actual economics, production KPIs, benefits, residual risks and accepted limitations. For every original requirement classify it as Delivered , Changed , Deferred , Rejected or Superseded , and provide rationale plus evidence. Do not copy earlier PRDs unchanged. Create docs/final-prd/ and save final-as-built-prd.md , requirement-disposition-matrix.md , final-capability-map.md , as-built-architecture.md , final-api-interface-baseline.md , final-data-knowledge-baseline.md , final-ai-agent-baseline.md , final-security-governance-baseline.md , final-nfr-baseline.md , final-acceptance-evidence-map.md , implemented-vs-deferred.md , known-limitations.md , residual-risk-register.md , production-kpi-baseline.md , final-economics-baseline.md , final-operating-boundaries.md ,	operational/economic/risk baseline

AI FDE Stage	Final Workshop Prompt — Mandatory Artifact Generation Included	Primary Evidence / Artifacts
	<code>final-ownership-map.md</code> , <code>approved-roadmap.md</code> and <code>final-product-baseline-signoff.md</code> . This becomes the authoritative baseline for operations, audit, onboarding, support and future change.	

Final Repository Spine

```
docs/
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│   ├── operating-contract/
│   └── ai-economics/
├── 01-engagement/
├── 02-stakeholders/
├── 03-problem-value/
├── 04-baseline-kpis/
├── 05-current-state/
├── 06-root-cause/
├── 07-repo-assessment/
├── 08-ai-qualification/
├── 09-initial-prd/
├── 10-architecture/
│   └── adrs/
├── 11-data-context/
├── 12-specs/
│   └── features/
├── 13-traceability/
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├── 15-modernization/
├── 16-repo-validation/
├── 17-implementation-prd/
├── 18-delivery/
├── 19-intelligence/
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├── 22-agentic-engineering/
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